THE COMMUTATOR IDEAL OF THE UNIVERSAL ASKEY-WILSON ALGEBRA EQUALS THE TWO-SIDED IDEAL GENERATED BY THE COMMUTATOR OF A AND B INSIDE THE SUBALGEBRA GENERATED BY A AND B

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Editorial note. The counted claim is the author Lemma lem:comu at source lines 3123-3131, and its proof is the proof environment that immediately follows it at source lines 3133-3168; the earlier lemmas are retained because the counted proof invokes them by label.

Editorial note. The spans at source lines 2467-2474 and 3009-3016 are retained so that the abbreviation O = <A,B> used by the counted statement is defined inside the material; the surrounding Section 10 discussion of the homomorphism from the q-Onsager algebra is not part of this task.

Editorial note. eq:caspreview is a display inside the introduction and is retained only because the definition of alpha, beta, gamma refers to it; it is not used as an input to any step of the counted proof.

Allowed standard prerequisites (editorial).

- Associative algebra and ideal conventions. Algebras are associative with a unit, subalgebras share that unit, and the two-sided ideal generated by a subset is the span of all products u s v with s in the subset.. Used in lem:three, lem:ideal and lem:comu at source lines 3049-3168.
- Commutativity criterion for a quotient algebra. If the images of a generating set commute in a quotient algebra, the quotient is commutative, so every commutator lies in the kernel.. Used at source lines 3066-3079 for the containment of the commutator subspace of O.
- Centrality of gamma. The element gamma of def:abc is central in Delta, as recorded in the retained definition at source lines 301-330.. Used at source lines 3094-3095, 3114-3117 and 3146-3150.

Our conventions for the paper are as follows. An algebra is meant to be associative and have a 1. A subalgebra has the same 1 as the parent algebra. We fix a field $\mathbb{F}$ and a nonzero $q \in \mathbb{F}$ such that $q^4 \neq 1$.

Definition 0.1. Define an $\mathbb{F}$-algebra $\Delta = \Delta_q$ by generators and relations in the following way. The generators are A, B, C. The relations assert that each of

$$A + \frac{qBC - q^{-1}CB}{q^2 - q^{-2}}, \qquad B + \frac{qCA - q^{-1}AC}{q^2 - q^{-2}}, \qquad C + \frac{qAB - q^{-1}BA}{q^2 - q^{-2}} \tag{0.1}$$

is central in Δ. We call Δ the *universal Askey–Wilson algebra*.

Definition 0.2. For the three central elements in (0.1), multiply each by $q + q^{-1}$ to get α, β, γ. Thus

$$A + \frac{qBC - q^{-1}CB}{q^2 - q^{-2}} = \frac{\alpha}{q + q^{-1}}, \tag{0.2}$$

$$B + \frac{qCA - q^{-1}AC}{q^2 - q^{-2}} = \frac{\beta}{q + q^{-1}}, \tag{0.3}$$

$$C + \frac{qAB - q^{-1}BA}{q^2 - q^{-2}} = \frac{\gamma}{q + q^{-1}}. \tag{0.4}$$

Note that each of α, β, γ is central in Δ. (The purpose of the factor $q + q^{-1}$ is to make the upcoming formula (0.5) more attractive.)

$$\Omega = qABC + q^2A^2 + q^{-2}B^2 + q^2C^2 - qA\alpha - q^{-1}B\beta - qC\gamma. \tag{0.5}$$

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Lemma 0.3. *The $\mathbb{F}$-algebra Δ is generated by A, B, γ. Moreover*

$$C = \frac{\gamma}{q+q^{-1}} - \frac{qAB - q^{-1}BA}{q^2 - q^{-2}}, \tag{0.6}$$

$$\alpha = \frac{B^2A - (q^2+q^{-2})BAB + AB^2 + (q^2-q^{-2})^2A + (q-q^{-1})^2B\gamma}{(q-q^{-1})(q^2-q^{-2})}, \tag{0.7}$$

$$\beta = \frac{A^2B - (q^2+q^{-2})ABA + BA^2 + (q^2-q^{-2})^2B + (q-q^{-1})^2A\gamma}{(q-q^{-1})(q^2-q^{-2})}. \tag{0.8}$$

Proof. Line (0.6) is from (0.4). To get (0.7), (0.8) eliminate C in (0.2), (0.3) using (0.6). □

Theorem 0.4. *The $\mathbb{F}$-algebra Δ has a presentation by generators A, B, γ and relations*

$$A^3B - [3]_q A^2BA + [3]_q ABA^2 - BA^3 = -\big(q^2-q^{-2}\big)^2(AB-BA),$$
$$B^3A - [3]_q B^2AB + [3]_q BAB^2 - AB^3 = -\big(q^2-q^{-2}\big)^2(BA-AB),$$
$$A^2B^2 - B^2A^2 + \big(q^2+q^{-2}\big)(BABA - ABAB) = -\big(q-q^{-1}\big)^2(AB-BA)\gamma,$$
$$\gamma A = A\gamma, \qquad \gamma B = B\gamma.$$

Proof. Use Lemma 0.3 to express the defining relations for Δ in terms of A, B, γ. □

Definition 0.5. Let $\phi : \mathcal{O} \to \Delta$ denote the $\mathbb{F}$-algebra homomorphism that sends $X \mapsto A$ and $Y \mapsto B$. Observe that ϕ is the restriction of φ to $\mathcal{O}$.

Definition 0.6. For notational convenience abbreviate $\mathbb{O} = \langle A, B\rangle$. Thus $\mathbb{O}$ is the image of $\mathcal{O}$ under the homomoprhism ϕ from Definition 0.5.

Lemma 0.7. $\Delta = \sum_{n=0}^{\infty} \mathbb{O}\gamma^n$.

Proof. The algebra Δ is generated by $\mathbb{O}$, γ. Moreover γ is central in Δ. □

Lemma 0.8. *The following (i)–(iii) hold.*

(*i*) $[\mathbb{O}, \mathbb{O}] \subseteq \mathbb{O}[A,B]\mathbb{O}$.
(*ii*) $[A,B]\gamma \in [\mathbb{O}, \mathbb{O}]$.
(*iii*) $\mathbb{O}[A,B]\mathbb{O}\gamma \subseteq \mathbb{O}[A,B]\mathbb{O}$.

Proof. (*i*) Abbreviate $R = \mathbb{O}[A,B]\mathbb{O}$ and consider the quotient algebra $\mathbb{O}/R$. The elements A, B generate $\mathbb{O}$, and these generators satisfy $[A,B] \in R$. Therefore $A+R$, $B+R$ generate $\mathbb{O}/R$, and these generators commute. This shows that $\mathbb{O}/R$ is commutative. Consider the canonical map $\mathbb{O} \to \mathbb{O}/R$. This map has kernel R. The map sends $[\mathbb{O}, \mathbb{O}] \mapsto 0$ since $\mathbb{O}/R$ is commutative. Therefore $[\mathbb{O}, \mathbb{O}] \subseteq R$.

(*ii*) In the third displayed equation of Theorem 0.4, the expression on the right is a nonzero scalar multiple of $[A,B]\gamma$. The expression on the left is equal to $[A^2, B^2] + (q^2+q^{-2})[B, ABA]$ and is therefore in $[\mathbb{O}, \mathbb{O}]$. The result follows.

(*iii*) By (*i*), (*ii*) above and since γ is central. □

Lemma 0.9. $\mathbb{O}[A,B]\mathbb{O}$ *is a 2-sided ideal of* Δ.

Proof. Abbreviate $R = \mathbb{O}[A,B]\mathbb{O}$. We show $\Delta R \subseteq R$ and $R\Delta \subseteq R$. Recall that Δ is generated by $\mathbb{O}$, γ. By construction $\mathbb{O}R \subseteq R$ and $R\mathbb{O} \subseteq R$. By Lemma 0.8(*iii*) and since γ is central we have $\gamma R \subseteq R$ and $R\gamma \subseteq R$. By these comments $\Delta R \subseteq R$ and $R\Delta \subseteq R$. □

Lemma 0.10. *We have*

$$\mathbb{O}[A,B]\mathbb{O} = \Delta[\Delta, \Delta]\Delta.$$

Proof. We have $\mathbb{O} \subseteq \Delta$ and $[A,B] \in [\Delta, \Delta]$ so $\mathbb{O}[A,B]\mathbb{O} \subseteq \Delta[\Delta,\Delta]\Delta$. We now show the reverse inclusion. To this end we analyze $[\Delta, \Delta]$ using Lemma 0.7. For integers $m, n \geq 0$,

$$[\mathbb{O}\gamma^m, \mathbb{O}\gamma^n] = [\mathbb{O}, \mathbb{O}]\gamma^{m+n} \overset{\text{by Lemma 0.8(}i\text{)}}{\subseteq} \mathbb{O}[A,B]\mathbb{O}\gamma^{m+n} \overset{\text{by Lemma 0.9}}{\subseteq} \mathbb{O}[A,B]\mathbb{O}.$$

By this and Lemma 0.7 we obtain $[\Delta, \Delta] \subseteq \mathbb{O}[A,B]\mathbb{O}$. Now using Lemma 0.9 we obtain $\Delta[\Delta,\Delta]\Delta \subseteq \mathbb{O}[A,B]\mathbb{O}$. The result follows. □